Activity 3A: Servo Motors

Purpose: In this activity, you will learn how to wire a servo and control it with code. Servo motors can be told to move to a specific position and stay there. Low-cost servo motors were originally used to steer RC airplanes and cars, but they have become popular for any project where precise movement is needed.

Materials:

Computer, Arduino Uno Board, USB cable, Breadboard, Potentiometer, Servo, Six jumper wires, Scissors

New Components:

Servo Motor: Servo motors have three wires: one for power, one for ground and one for signal. When you send the right signal through the signal wire, the servo will move to a specific angle and stay there. Common servos rotate over a range of about 0° to 180°.



BROWN is GROUND
RED is POSITIVE
ORANGE is DATA (assigned to a pin on the Arduino)

Note: Its best to avoid driving the servo all the way to 0 or 180 degrees. It can cause the motor to jitter, which is bad for the servo. So typically a range of 20-170 works best.

Servo Motor Components: Included with your servo motor you will find a variety of motor mounts that connect to the shaft of your servo. You may choose to attach any mount you wish for this circuit.

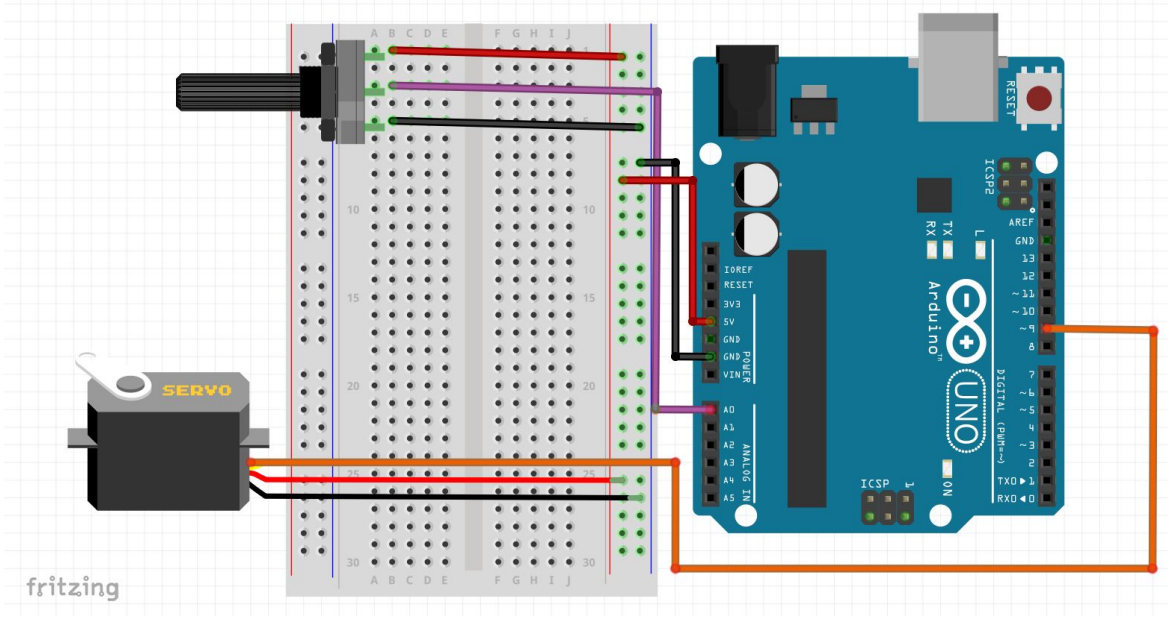


Potentiometer (pot): A potentiometer is a 3 pin variable resistor. The middle pin outputs a voltage between 0 and 5 volts depending on the position of the knob. The more volts that the pot allows to travel through the circuit, the greater the volume, speed, or intensity of the function will be. We need to hook up the pot to one of the analog pins which are labelled A0 to A5. The inside post of the “pot” needs to be hooked to an analog pin.



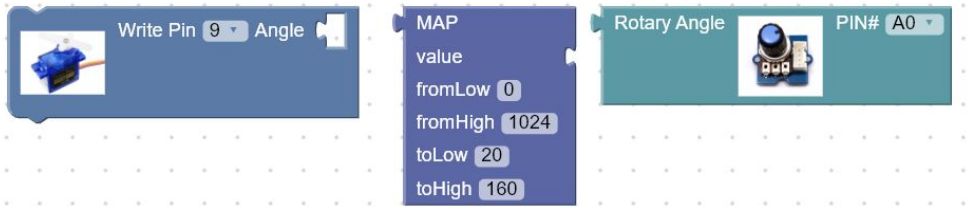
Procedure:

1. Wire the project using the following schematic.



Code

You can create your code in the BlocklyDuino then paste it to the Codebender website. Click BlocklyDuino Reference for a link to all the blockly codes. Here are the blocks you will need.

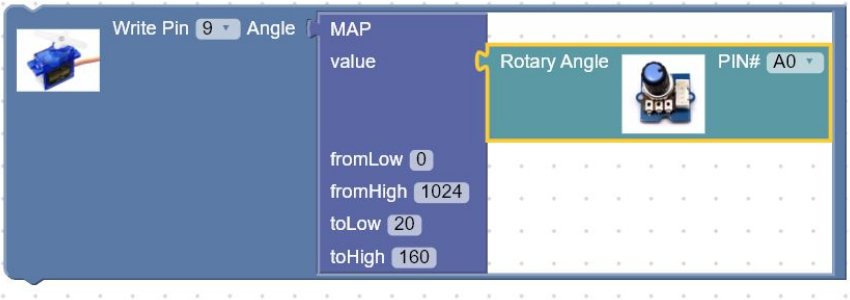


1. Select the servo from the blocks menu and assign it to pin 9. Servo has its own set of blocks and they are found under “SERVO” on the left hand side of the menu. We are then going to assign a map function to the angle in the next step.

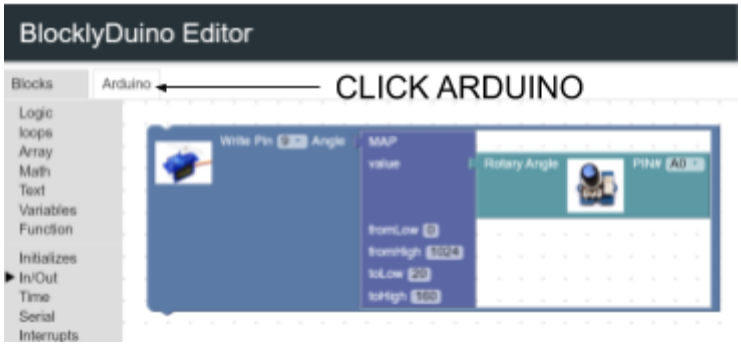
2. The code below uses the “MAP” function. The MAP function converts the resistance of the potentiometer (0-1024) to an angle (0-180). You will find the map block in “MATH.” Drag the block into the blockly editor and attach it to the angle of the servo. Therefore, the angle of the servo will depend on a value that the potentiometer gives. We will add the “pot’ with the next step. 7

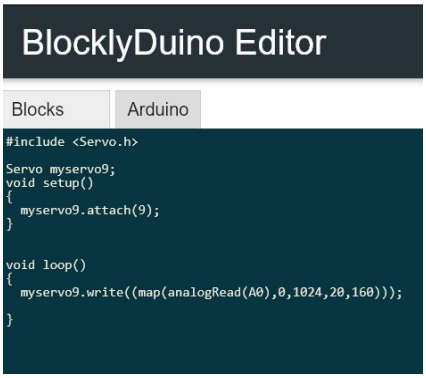
3. The potentiometer (rotary angle) controls the resistance. Select “rotary angle” from the GROVE menu under “INPUTS” and assign it to pin A0

4. If the servo motor is jittery when you moving it, switch your angles in your code from (0-180) to (20-160). Your code should look like this:



5. Once your blockly code is finished, click on the arduino to transfer the information in the blocks to Arduino.





*****SAVE your sketch to your BlocklyDuino Folder that you created*****

You can now copy the Arduino code by using “ctrl A” then “ctrl C”. Then paste this into the Codebender website. Click the arrow to upload the following sketch onto your board.

What You Should See:

Turning the potentiometer will cause the servo arm to turn. The servo will mimic the movement of the potentiometer, twisting in the same clockwise or counterclockwise direction. If you’ve attached a servo mount to the arm as shown, this movement will be easier to see.

Troubleshooting: If your circuit is not working!

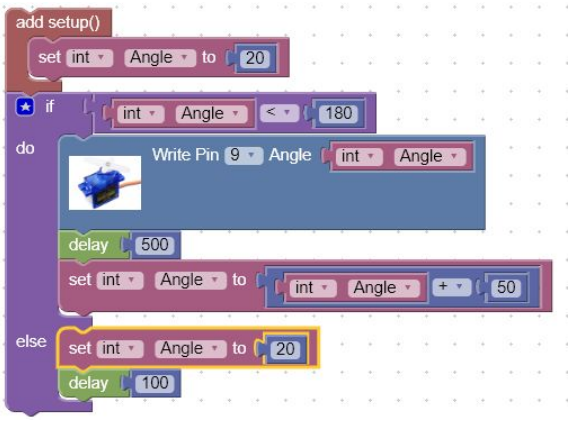
1. Double check your code! Did you put in every block? Are the settings correct?
2. Refresh the website and reload the code.
3. The next step is to check your circuit!
    a. Check your wiring! Is it the same as the circuit diagram? Does the circuit go from the positive numbered pin to the ground?
    b. Next check each component. Are they wired to the correct terminal?

c. Check the speaker. Rewire it to pin A0. Open a new Blocklyduino window and test the speaker by playing this code:



This should cause the speaker to chirp.

d. Check the servo. This can be achieved by connecting it to pin 9 and running a test program. Open a new window of Blocklyduino and load the below code.



The servo should now move from 20 to 70 to 120 to 170 degrees and back to the start position.

e. Next check the individual wires in the circuit. This can be done by connecting an LED to pin 13 only then using the wire to ground the short pin. Do not connect the pin to the ground directly. (Shorting the wires) This will damage the UNO.

f. For further information on using the serial monitor see Activity 4A Using a Serial Monitor Display.

Assignment 3A SERVO

Attach a servo mount to the arm so that you can observe the motion of the servo motor to answer the following questions:

1. What did the servo do when you turned the potentiometer clockwise? Counter clockwise?

2. How would you get the servo to rotate at a different angle?

3. What are some real life applications would incorporate a servo and a potentiometer?

CHALLENGE

4. Servos can be controlled using code to move at different degrees without the use of a potentiometer. Copy and paste the following code to operate a servo without a pot.
    - Remove the potentiometer from your wiring. You will now code your arduino to operate a servo using code (automation).

    - Now program your arduino using the following blocks.



a. What did your servo do?

b. Change your code to make your servo rotate 170 degrees, wait for ½ a second and rotate to 90 degrees, wait for 2 seconds and rotate back to 10 degrees.

c. Add a loop so that the servo repeats the code 10 times.

Note: Its best to avoid driving the servo all the way to 0 or 180 degrees. It can cause the motor to jitter, which is bad for the servo. So typically a range of 20-170 works best.

5. Code and wire two servos to work in succession. Have one servo turn to 90 degrees. Input a wait time so that another servo can turn to 120 degrees and then back to 0 degrees. Then have the first servo return back to 0 degrees. (“Dancing Servos:)”